

DesignSpark: For All Your PCB Designing Needs

DesignSpark PCB is a schematic and layout design software. In this article we explore the capabilities of this software, highlight the new features that came in its latest version, and also shed light on DesignSpark Mechanical, which is another tool included in the DVD accompanying EFY Plus

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DesignSpark PCB is a powerful, professional and completely free schematic and layout design software. Ideal for use in professional design environment, it currently claims over 250,000 users worldwide. DesignSpark PCB is easy to learn and comes with an intuitive user interface. It features context-sensitive toolbars, context-sensitive help, customisable shortcuts, interaction bar, easy access

to all relevant commands and an online library of components—ModelSource—with some reference designs.

DesignSpark PCB also has online video and text tutorials to help users get started and a dedicated technical forum for technical clarifications. It improves itself based on community feedback, and further software development, enhancements and new features are derived from comments, design work and experiences shared by other members at DesignSpark's forum.

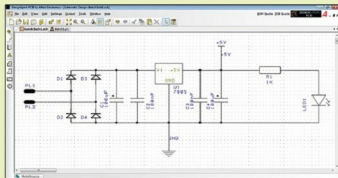


Fig. 1: User interface of DesignSpark PCB

sign and layout, generating impressive 3D view to visualise your design in real time and generate manufacturing files.

Compatibility

DesignSpark PCB supports importing of CadSoft Eagle PCB CAD design files, circuit diagrams and libraries. In addition to extensive libraries, sophisticated part-creation wizards make it easy to design new parts from scratch or by amending downloaded symbols and footprints. Bill of materials (BOM) report can also be generated at any time.

DesignSpark PCB allows you to generate all the files required to take your project from circuit diagram to layout schematic to a circuit-board built project. Gerbers (extended RS-274-X), excellons and component position reports are available to make your design compatible with pick-and-place machines ready for automated PCB assembly (reflow oven process). Other output formats include DXF (import and export) and IDF for interfacing to mechanical CAD software tools.

Key features

Let us take a look at some of the key features of this tool:

Reference designs. DesignSpark PCB has reference designs for hardware evaluation modules and development kits with complete schematics, PCB files and detailed documentations. Designers can save a lot of their time by using these designs and modifying them according to their requirements.

Additions in version 6 (in this month's DVD)

- 1. Easy IDF export to DesignSpark Mechanical.** DesignSpark PCB version 6 has simplified IDF export to DesignSpark Mechanical by removing the need to do any configuration setting. Fully configurable IDF export is available to export in other MCAD tools.
- 2. Cross Probe.** The new Cross Probe (X-probe) functionality further speeds up the design process by allowing instant referencing between schematic and PCB design views within the software.
- 3. Custom Shortcuts.** DesignSpark Mechanical version 6 lets designers define their custom shortcuts in the software. This flexibility to designers speeds up the entire design process.
- 4. Start Page.** As soon as designers open DesignSpark PCB they see lists of useful electronics highlights, reference designs, getting started links and the handy list of recent projects/files.